

NLP and LLM Applications

Natural language processing concepts for modern AI applications

1. Natural Language Processing

Natural Language Processing (NLP) focuses on enabling computers to process, analyze, and generate human language. Tasks include classification, sentiment analysis, named-entity recognition, information extraction, summarization, and question answering.

2. Text Representation

Traditional NLP approaches include bag-of-words and TF-IDF. Modern systems frequently use dense vector representations called embeddings. Embeddings encode semantic relationships and can support similarity search and retrieval.

3. Transformers

Transformer architectures use attention mechanisms to model relationships between tokens. They became a foundation for many modern language models because they can process contextual relationships efficiently and scale to large datasets.

4. Large Language Models

Large Language Models (LLMs) generate text by predicting tokens based on context. They can perform summarization, question answering, extraction, rewriting, coding assistance, and conversational tasks. Their outputs should still be validated for factual accuracy.

5. Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) combines information retrieval with generation. Relevant passages are retrieved from a knowledge source and supplied to the model as context. This can help an application answer questions about a specific document collection.

6. Useful Python Tools

Common tools include spaCy for NLP pipelines, Hugging Face Transformers for pretrained models, scikit-learn for classical machine learning, and vector databases or similarity-search libraries for retrieval-based applications.

This document is provided as downloadable project/reference material.